UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: COMPUTER NETWROKS

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SECTION : A

SEMESTER : 4th

SUBMITTED T0 : NAJEEB SIR

Lab 5

Computer Networking Lab
Using Python Requests Library to experiment with some aspects of HTTP.

Introduction

In this lab, you will use python's request library to study various aspects of HTTP. Requests is a HTTP library for the Python programming language.

You should already have python installed on your system.

To install the requests library, use the following commands:

```
pip install requests
```

Lab Tasks

Using requests library, access a website of your choice and then extract all header fields one by one and print them in a clean way

```
import requests
r = requests.get('https://web.facebook.com/')
print(r.status_code)
```

```
200
```

```
import requests
r = requests.get('https://www.booking.com/')
print(r.status_code, r.reason)
```

```
200 OK
```

```python
import requests
r = requests.get('https://www.booking.com/')
print(r.status_code, r.reason)
```

```
200 OK
```

```python
import requests
r = requests.get('https://uetmardan.edu.pk/tours')
print(r.status_code, r.reason)
```

```
404 Not Found
```

```python
import requests
r = requests.get('https://uetmardan.edu.pk')
print(r.headers)
```

```
{'Date': 'Wed, 05 Mar 2025 07:13:07 GMT', 'Server': 'Apache', 'X-Powered-By': 'PHP/8.1.31', 'Expires': 'Thu, 19 Nov 1981 08:52:00 GMT', 'Cache-Control':
```

```python
import requests

r = requests.get('https://www.booking.com/')

print("Accept-Ranges:", r.headers.get('Accept-Ranges'))
print("Cache-Control:", r.headers.get('Cache-Control'))
print("Connection:", r.headers.get('Connection'))
print("Content-Encoding:", r.headers.get('Content-Encoding'))
print("Content-Length:", r.headers.get('Content-Length'))
print("Content-Type:", r.headers.get('Content-Type'))
print("Date:", r.headers.get('Date'))
print("ETag:", r.headers.get('ETag'))
print("Expires:", r.headers.get('Expires'))
print("Keep-Alive:", r.headers.get('Keep-Alive'))
print("Last-Modified:", r.headers.get('Last-Modified'))
print("Server:", r.headers.get('Server'))
print("Transfer-Encoding:", r.headers.get('Transfer-Encoding'))
print("Vary:", r.headers.get('Vary'))
print("X-Content-Type-Options:", r.headers.get('X-Content-Type-Options'))
print("X-Frame-Options:", r.headers.get('X-Frame-Options'))
print("X-XSS-Protection:", r.headers.get('X-XSS-Protection'))
```

```
Accept-Ranges: None
Cache-Control: private
Connection: keep-alive
Content-Encoding: gzip
Content-Length: 190431
Content-Type: text/html; charset=UTF-8
Date: Wed, 05 Mar 2025 07:07:44 GMT
ETag: None
Expires: None
Keep-Alive: None
Last-Modified: None
Server: nginx
Transfer-Encoding: None
Vary: Accept-Encoding, User-Agent
X-Content-Type-Options: nosniff
X-Frame-Options: None
X-XSS-Protection: 1; mode=block
```